

Lab-I
National Institute of Technology Silchar

Subject Code: CS-5104
Semester: 1st M.Tech.
Section: All

Subject: Computer System Lab
Department: CSE
Date: 07 September 2026

Teaching Assistant (TA) will ensure that you have completed your tasks or not. The TA will record the performance for future grading. You need to write the programs and/or outputs on your lab copy as directed in the questions.

1. Write a C program using `fork()` to create three processes that perform different operations on an integer array. The parent process calculates the sum of all array elements, the first child process finds the maximum element, and the second child process finds the minimum element.
2. Write a C program to implement two-way communication between a parent and child using two pipes.
3. Write a C program to simulate the First Come First Serve (FCFS) CPU scheduling algorithm and calculate the waiting time and turnaround time of each process.
4. Write a C program to simulate the non-preemptive Shortest Job First (SJF) CPU scheduling algorithm and calculate the waiting time and turnaround time of each process.
5. Write a C program to simulate the Round Robin CPU Scheduling Algorithm for a given time quantum. Display the processing sequence, burst time, waiting time, and turnaround time of each process, along with the average waiting time and average turnaround time.

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